Remote Sensing Semantic Segmentation

Four-Model Development → Partial Cross Entropy

Technical Report — visual summary of the experimental progression

1. Experiment at a Glance

Stage	Purpose	Mean IoU	Mean Dice
Model 1	Initial baseline	15.45%	22.01%
Model 2	Class-balanced model	24.03%	32.96%
Model 3 ★	Best fully supervised baseline	55.74%	66.27%
Model 4	Partial CE — 100 points/image	53.31%	62.93%

Core question
Can useful segmentation be retained when full pixel-level supervision is replaced by extremely sparse point labels?

2. Why the Models Evolved

Transition	What the results showed	Decision
Model 1 → 2	Model 1 was weak overall; several classes were barely learned.	Introduce class-balanced training to address class imbalance.
Model 2 → 3	Class balancing improved results, but performance remained inadequate.	Develop/select a substantially stronger fully supervised setup as the reference baseline.
Model 3 → 4	The best baseline established a strong reference.	Change the supervision regime: test Partial CE with only 100 labeled points/image.

Development logic
Weak baseline → address imbalance → strongest full-mask baseline → test whether sparse supervision can approach that baseline.

3. Early-Model Evidence

Class	Model 1 IoU	Model 2 IoU
Urban	24.46%	33.49%
Agriculture	63.00%	66.70%
Rangeland	0.00%	1.05%
Forest	9.34%	49.17%
Water	11.34%	11.44%
Barren	0.00%	6.36%
Unknown	0.00%	0.00%

Model 2 improved Mean IoU from 15.45% to 24.03% and Mean Dice from 22.01% to 32.96%. Forest showed a particularly large improvement, but Rangeland, Barren and Unknown remained difficult.

4. Best Baseline vs Partial CE

Metric	Model 3 — Full mask	Model 4 — Partial CE	Change
Mean IoU	55.74%	53.31%	−2.43 pp
Mean Dice	66.27%	62.93%	−3.34 pp
Supervision	Full mask	100 points/image	≈0.15% pixels labeled

Main finding
Partial CE retained 53.31% Mean IoU and 62.93% Mean Dice using only 100 labeled points per 256×256 image. The Mean IoU decrease from the full-mask baseline was only 2.43 percentage points.

5. Partial CE — Per-Class Performance

Class	IoU	Dice
Urban	64.19%	78.19%
Agriculture	82.44%	90.38%
Rangeland	13.49%	23.78%
Forest	78.48%	87.94%
Water	75.85%	86.26%
Barren	58.70%	73.98%
Unknown	0.00%	0.00%
Mean	53.31%	62.93%

6. Predicted Class Distribution

Class	Model 3	Partial CE
Urban	9.817%	9.335%
Agriculture	52.894%	63.531%
Rangeland	10.641%	2.414%
Forest	12.206%	10.307%
Water	5.006%	4.449%
Barren	9.436%	9.964%
Unknown	0.000%	0.000%

Interpretation

Partial CE shifts predictions toward Agriculture and away from Rangeland. This agrees with the per-class IoU results: Rangeland is the main class affected by sparse supervision.

7. Experimental Process

Step	Process
01	Train and evaluate the three fully supervised model stages; select Model 3 as the strongest reference.
02	Randomly sample 100 labeled pixels per image to simulate sparse point annotation.
03	Train the segmentation model using Partial Cross Entropy only at labeled pixel locations.
04	Evaluate predictions against the complete validation masks using Mean IoU, Dice and per-class scores.

8. Qualitative Results

The validation visualizations compare: **satellite image → complete ground truth → Partial CE prediction**

The predictions recover major spatial regions, while some fine structures and difficult classes are less accurately represented. The strongest quantitative weakness is Rangeland.

9. Conclusion

Conclusion

The experiments progress from a weak initial model (15.45% mIoU), through class balancing (24.03%), to a substantially stronger fully supervised baseline (55.74%). Partial Cross Entropy then achieved 53.31% mIoU with only 100 labeled points per image (≈0.15% of pixels), showing that sparse supervision can retain much of the baseline segmentation quality. The main limitation is uneven class performance, especially for Rangeland.

55.74% full-mask mIoU → 53.31% Partial-CE mIoU | 100 points/image | ≈0.15% labeled pixels